Multi-Disease Detection Report

CNN + LLM contextual reasoning • generated 2026-08-24 23:02

Patient context

Age	72
Sex	Female
Symptoms	Short-term memory loss, repeatedly asking the same questions, forgetting recent conversations, difficulty finding words
History	72-year-old woman, gradual memory decline over 2 years, increasingly relies on family members for reminders

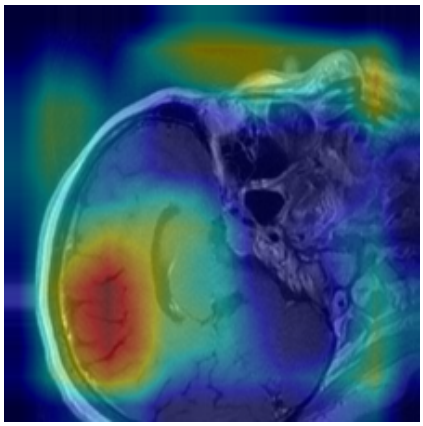
Model prediction

brain_pituitary **Malignant**

Brain Tumor · Neurological / Brain · Brain MRI

#	Condition	Confidence
1	brain_pituitary	83.0%
2	brain_meningioma	12.4%
3	kidney_tumor	1.1%
4	cervix_dyskeratotic	0.6%
5	lymph_mantle_cell_lymphoma	0.6%

Grad-CAM (model attention)



LLM contextual interpretation

Model suggests Brain Tumor (brain_pituitary) at 83.0% confidence.

Model finding. The CNN's top class is 'brain_pituitary' (Malignant, Neurological / Brain) on Brain MRI, at 83.0%.

Confidence. Moderate confidence (83.0%); margin to runner-up is 70.6 pts — consider the differential.

Differential

- **brain_pituitary** — ranked #1 at 83.0%
- **brain_meningioma** — ranked #2 at 12.4%
- **kidney_tumor** — ranked #3 at 1.1%

Grad-CAM correlation. Model attention was a moderately sized region in the central-left portion of the image (~18% high-salience area). Verify this region is anatomically plausible for Brain MRI.

Biomedical context. [KB] Pituitary tumor affecting the sellar region / hormonal axis. Group: Brain Tumor; organ: Neurological / Brain; modality: Brain MRI; category: Malignant (severity: moderate). Primary brain tumors include gliomas, meningiomas, and pituitary adenomas, differing in location, growth pattern, and MRI appearance; imaging plus histology establish the diagnosis. [BioGPT fallback — set ENABLE_BIOGPT=true to run the real model] MRI with contrast characterizes the lesion; definitive typing requires histopathology/molecular markers, often after surgical sampling. [Med-PaLM 2: PLACEHOLDER — no public API; see medpalm_adapter.py to wire MedLM]

Patient considerations. Provided context (age: 72; sex: Female; symptoms: Short-term memory loss, repeatedly asking the same questions, forgetting recent conversations, difficulty finding words; history: 72-year-old woman, gradual memory decline over 2 years, increasingly relies on family members for reminders) should be correlated with the finding.

Recommended next steps

- Correlate with the full clinical picture and prior imaging.
- Obtain gold-standard confirmation appropriate to the modality (e.g., biopsy/histology or specialist review).
- Escalate to the relevant specialist if the category is malignant or high-severity.

Uncertainty & limitations

- Interpretation generated WITHOUT the Claude reasoning model (offline fallback).
- Model trained on specific public datasets; performance may not transfer to other scanners/populations.
- A confident softmax score is not calibrated probability of disease.

■ Research decision-support only — not a medical diagnosis. All findings require confirmation by a licensed clinician.